Crossing the Coordination Valley: Hierarchical Managers over Options for Two-Front Tactics in a Drone-Based Predator–Defense Game

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***Abstract*—We study the emergence of coordinated two-front tactics in a 2D simulation of livestock protection: a wolf pack attacks a herd defended by drones. An effective non-lethal defense yields a double benefit: ranchers stop losing livestock and, once the damage disappears, so does the main incentive for lethal predator control, reducing retaliatory wolf killings and fostering coexistence. The attacker's key tactic—the *bait*, in which one wolf exposes itself to drag the defense away while the pack strikes the opposite flank—requires crossing a reward valley: splitting the pack worsens the return for thousands of steps before the deception pays off. We show empirically that flat reinforcement learning does not cross this valley on either side, and that residual variants initialized at the scripted tactic *erode* it. We propose a minimalist hierarchical architecture—a PPO manager choosing among frozen scripted options, formulated as an event-terminated semi-MDP with a deliberation cost—and apply it symmetrically to both sides. The attacking manager converges in 3 out of 3 independent trainings to the same policy in the hard stratum ($P(\Delta 90 \,|\, S) = 1.000$), rediscovering from pure reward the manually distilled strategy, with non-inferiority to the expert oracle reproduced in replication. The defending manager learns to split its forces when a second front appears and reduces losses relative to the classical barrier by a factor of 2.5–3.5 depending on the condition (up to 3.5$\times$ on the natural mix). As a methodological result, we document three episodes of *specification gaming* caught by pre-registration and human behavioral auditing.**

***Index Terms*—multi-agent reinforcement learning, hierarchical learning, options, semi-MDP, coordination, specification gaming, drone swarms, predator coexistence**

I. Introduction

Wolf predation on livestock is a persistent conflict wherever the two coexist, and its usual management—after-the-fact compensation and lethal predator control—is costly for the rancher and lethal for the wolf. An active *non-lethal* drone-based defense attacks the problem from both ends: if the herd suffers no losses, the rancher loses nothing and the incentive for retaliation disappears. This work builds on a validated 2D simulation of that scenario and on a prior result of the same project: a defense learned by reinforcement learning (MARL) outperforms a classical rule-based barrier at fine-grained piloting (0.77 vs. 0.93 head of livestock lost per episode). The central question is whether the *two-front* tactical coordination—the attacker's bait and the defender's force splitting—can be learned at all, or lies beyond the reach of flat RL.

The tactic of interest is the **two-front bait** (Fig. 1): a lone wolf lets itself be seen on one flank to draw the drone barrier, while the pack attacks the opposite, uncovered flank. It is role coordination with sacrifice: the decoy gives up hunting so that the others may kill. Our hypothesis, confirmed by the experiments, is that this tactic is separated from greedy behavior by a **reward valley**: splitting the pack reduces short-term kills (the hunting quorum is lost on each front) for thousands of simulation steps before the deception pays off. An optimizer improving step by step reaches the valley's edge, observes that splitting makes things worse, and turns back: the pathology known as *relative overgeneralization* [1], [2], here in its temporal variant, since it persists even under a centralized team policy.

Our contributions are fourfold:

1) **Diagnosis**: systematic evidence that flat RL does not cross the valley on either side: eight attack campaigns (pure PPO, potential-based shaping [3], behavior-cloning warm starts, residual learning on top of the script [4], and curricula) fail to make the bait emerge, and the residual ones *erode* it; on the defending side, 20M steps of MAPPO with full movement freedom produce no front splitting (Sec. V-A).
2) **Architecture**: a minimalist hierarchical manager—PPO over a discrete space of frozen scripted options, an event-terminated semi-MDP with a deliberation cost—implementable on Stable-Baselines3 [5] with no additional libraries (Sec. IV).
3) **Symmetric results**: the attacking manager converges to the same optimum in 3/3 independent trainings and is non-inferior to the expert oracle (reproduced in replication); the defending manager learns conditional force splitting and cuts the classical barrier's losses by a factor of 2.5–3.5 (Sec. V).
4) **Methodology**: a protocol of pre-registration, paired seeds and human behavioral auditing that detected and closed three episodes of *specification gaming* [6], [7] and

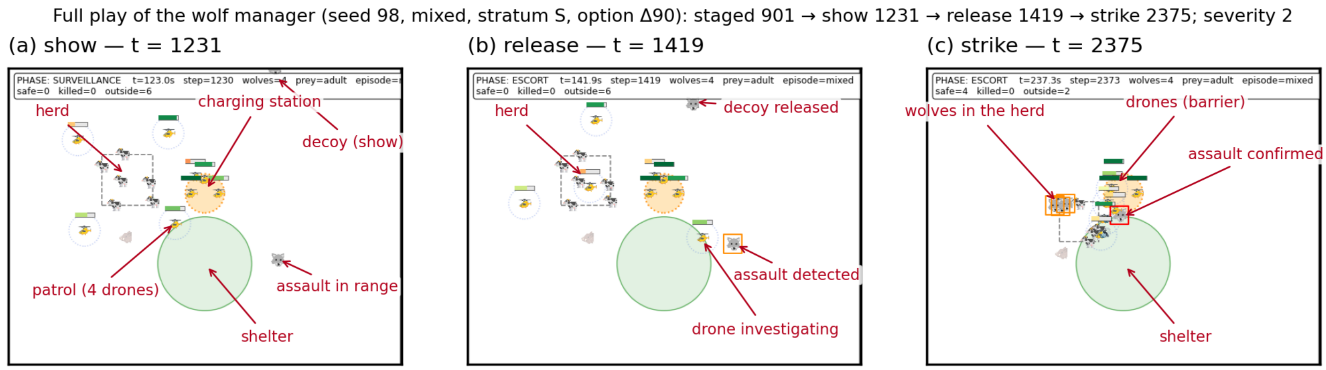

Fig. 1. The complete play learned by the attacking manager (seed 98, dispersed stratum, a single sustained $\Delta 90$ option): (a) the decoy shows itself and the barrier anchors to it (t=1231); (b) the assault is released on the opposite flank (t=1419); (c) the strike (t=2375). One high-level decision executed from start to finish.

five environment defects before they could contaminate the conclusions (Secs. III-E and V-D).

II. Related Work

The coordination valley. *Relative overgeneralization*—converging to suboptimal equilibria because coordinated actions look worse under teammates' current behavior—has been attacked from three families. *Directed exploration* uses joint novelty to force visits to the far side of the valley (JIM [8], EMC [9]). *Optimistic or lenient updates* modify the learning rule so that cooperative actions are not punished prematurely [2], [10]. The third family, to which this work belongs, *hierarchically restructures* the decision space so that the coordinated maneuver costs one discrete choice instead of a thousand-step choreography: the first two families *wade* the valley; the hierarchy *bridges* it.

Hierarchy and allocation in MARL. The options/semi-MDP framework is classical [11]; learning options end-to-end (Option-Critic [12]) suffers from option degeneracy, and manager–worker hierarchies with continuous goals (FeUdal [13], HIRO [14]) add unnecessary machinery when roles are discrete and nameable. Closest to our design are low-frequency centralized allocators over low-level policies: COPA [15], ALMA [16] and HMASD [17]; and, in swarm confrontation, discrete allocation layers decoupled from continuous execution [18]. Our contribution relative to this line is not the allocator–executor pattern, but its explicit use to cross a reward valley with delayed payoff—an *anti-greedy* allocation—and its symmetric demonstration on attacker and defender of the same environment, with replications and pre-registration.

Domain. The pack's hunting behavior is inspired by the simple-rules model of Muro *et al.* [19]; robotic shepherding builds on Strömbom's heuristic [20], and aerial shepherding with hierarchical RL was explored by Abbass's group [21]. In pursuit–evasion, flat MARL does discover spatial coordination (flanking, splitting) when the reward is dense and monotone [22]; our result delimits that frontier: flat learning splits when splitting is monotonically good, and fails when it requires getting worse first.

III. The System

A. World

The simulation (NumPy, frozen and versioned physics) takes place on a 500×500 m field with timestep $dt = 0.1$ s and episodes of up to 23,570 ticks[1] (Fig. 2). It features 4 active drones plus 4 reserves with battery-driven relays, 6 adult cows with 0–2 calves, and 1–5 wolves per episode entering from the perimeter in one or two angularly separated groups ($\geq 60°$). Maximum speeds: drone 15 m/s, wolf 4 m/s, cow 1.2 m/s. An adult dies when ≥ 2 simultaneous wolves flank it at ≤ 3 m outside its frontal cone; a calf, to a single wolf that evades its guardian cow.

B. Asymmetric perception and deterrence

Wolves are omniscient (adversary model); the defense only knows what it perceives: *detection* at ≤ 100 m from any active drone and type *confirmation* at ≤ 40 m, with team memory. Deterrence is acoustic: any wolf within 20 m of an active drone flees. This asymmetry is what makes the bait possible: the classical barrier reacts only to confirmed threats and anchors its line to the first confirmation, leaving the opposite flank uncovered.

C. Defenses and scripted attack

The *classical barrier* patrols a static ring around the herd and, upon confirmation, deploys a rigid line of slots 20 m apart facing the anchor. The *scripted bait* implements the two-front tactic by hand (decoy loitering out of reach, dark approach of the assault, synchronized trigger) and serves as the attack's upper reference. As an intermediate defensive rule we use a *proportional allocator*: upon a second threat cluster, it detaches $\min(n_{\text{wolves}}, 2)$ guards and keeps the rest in line.

[1]$\approx$39 simulated minutes at $dt = 0.1$ s; a cap inherited from the original calibration and kept frozen for cross-version comparability.

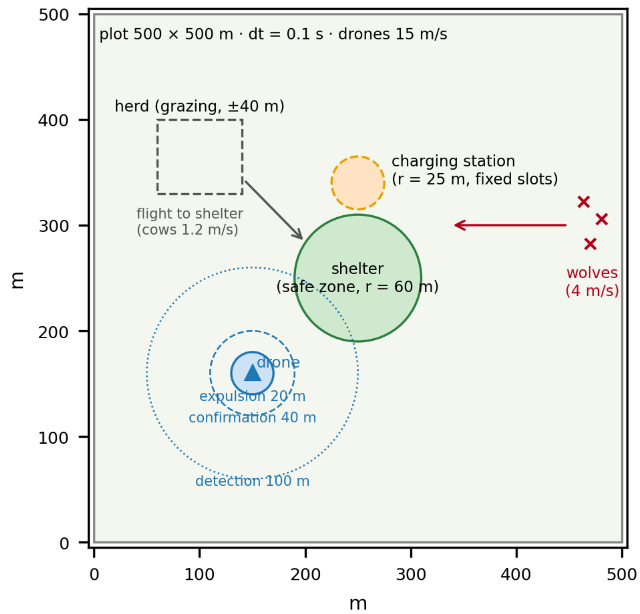

Fig. 2. Environment schematic: 500 × 500 m field with central shelter, charging station, herd, and the drone's perception/deterrence radii (expulsion 20 m, confirmation 40 m, detection 100 m). Positions are illustrative.

D. Relays, station and deliberate gaps

Battery autonomy forces periodic relays with the reserve. The final protocol is a *sentinel* one: every formation post and every charging bay is a fixed position; the outgoing drone signals and stays pinned at its post (stationary deters just as well), the incoming one flies straight to those coordinates and the hand-off occurs at ≤ 2 m; the outgoing drone then returns to the bay its relief left reserved. "One leaves, one enters, same position": no other drone relocates. On patrol, the ring adapts its radius to the herd's spread ($\approx$ 69–97 m), which with four units fixes neighbor separation at $D = 2R\sin(\pi/4) \approx 1.41R$: strict surveillance coverage ($D \leq 100$ m) is reached only at the minimum of the observed spread range ($R \lesssim 71$ m) and is unattainable elsewhere, but the hard rule of opening no blind corridors ($D \leq 2\,r_{\text{det}} = 200$ m) holds throughout the audited corpus. The remaining gaps are *deliberate*: the line covers the anchor's front and leaves flanks and rear open—the bait's target—and the *investigation reflex* (upon a detection, the nearest free drone goes to look) temporarily removes a drone from the ring. The corpus's only episode with undetected entries combined both: a lone wolf dragged the investigator away, the pack entered through the weakened arc and the escort latched 17 ticks too late into a total loss; the phenomenon disappeared under v3.7, but it illustrates why force allocation is the critical defensive decision.

E. An audit-driven engineering chronicle

The environment underwent four revisions (v3.4→v3.7), all originating in human viewing of rendered episodes and subsequently converted into regression tests; we highlight them because each one altered conclusions that automated tests deemed sound:

1) *The sound rule* (v3.5). Viewing revealed wolves crossing the barrier through the midpoint between two drones, where repulsion forces canceled out and expulsion—conditioned on the drone approaching—never triggered. Deterrence was redefined as an always-on field (≤ 20 m from any drone). The consequence was major: the previous MARL defense's advantage evaporated after the fix (it had partly learned to exploit the defect), and the entire comparison ladder was re-measured and re-trained.
2) *Static patrol* (v3.6). Relays against orbiting posts failed often (chasing a moving hand-off point); with always-on sound, a stationary drone deters just as well, so orbiting no longer contributed anything. Relay waiting time dropped from 298 to 102 ticks and stranded drones to zero.
3) *Sentinel relay* (v3.7). Fixing field posts and charging bays ("one leaves, one enters, same position") uncovered that the patrol had been silently rotating every drone one slot per relay since v3.0—unordered motion on which, unknowingly, guarantees of the attack script rested; they had to be reformulated as best-effort semantics. Distilled principle: no behavior may rest on unordered motion.
4) *Coverage audit*. The resulting design rule—no two formation neighbors may open an uncovered corridor—was formalized ($D \leq 2r$; with M drones on a ring of radius R, $D = 2R\sin(\pi/M)$, so the strict threshold would require $M{=}6$), and its auditor caught the corpus's only undetected-entry mode: the investigation reflex removes a drone from the ring and opens the neighboring arc, a resource-allocation defect of the classical defense that disappeared under v3.7.

Every correction was versioned, re-measured and re-frozen before continuing (Sec. IV-D); the simulator's bitwise determinism allowed each change to be verified with paired seeds.

IV. Method

A. Flat baseline

On the defending side, MAPPO [23] with shared actors and a centralized critic (CTDE), continuous residual action on top of the barrier [4]. On the attacking side, single-brain PPO [24] (one policy moves all 5 wolves), in variants: pure sparse reward, potential-based shaping [3], behavior-cloning warm start, and residual learning on the script with and without an angular separation curriculum and oversampling of two-front episodes.

B. Hierarchical architecture

Options layer. Existing scripted behaviors are encapsulated as parameterized, frozen options. Attacker: MASS (greedy hunt) and BAIT(Δ)—decoy plus assault along the current heading (*keep*) or shifted 90° / 180°—with hunting rules uniform across all arms (prey persistence, re-selection only when the prey is guarded, hysteresis and a cooldown). Defender: deployments {4-0, 3-1, 2-2} splitting the four posts between the line and guards that chase the secondary cluster. The BAIT option encapsulates the full choreography as a state machine: decoy designation (the lowest-index wolf), dark approach of the assault along an outer ring of ≈ 150 m, decoy loitering out of reach (> 130 m from the nearest drone), a "broken-wing" display, waiting for the barrier's commitment, and a synchronized release; a deterministic matcher translates the manager's decision into memberships, so the high-level choice is genuinely single-step.

Semi-MDP manager. One manager per side (PPO, $[64, 64]$ network, episodic $\gamma = 1$) observes a permutation-invariant tactical summary of $\sim$35–40 features (threat clusters and their angular separation, per-octant openness, deception-progress features, herd state and clock; the defending manager sees only contacts and confirmations) and picks an option. The option retains control until a *terminal event* (a death, strike resolution, herd safe, pre-trigger abort, or a 4,500-tick cap): event-based termination instead of a fixed clock. Every voluntary interruption of a running option pays a *deliberation cost* $\varepsilon = 0.05$ in reward, which penalizes strategic churn without forbidding any decision: the manager keeps full freedom and must earn every change of plan. For the attacker, the observation includes own structure (alive count, number of threat clusters and their angular separation—the feature that separates the strata), the defense (per-octant "gate" distance, fraction of active drones inside the decoy's cone and the barrier's projected displacement: the continuous ingredients of deception progress), herd state and the escort clock; for the defender, clusters of contacts$\cup$confirmations with per-octant temporal memory and the current own deployment. Training uses 120k macro-decisions with 24 parallel environments (lr $3 \cdot 10^{-4}$, entropy annealed $0.02 \to 0.005$) and costs hours of CPU: an order of magnitude less than a flat campaign.

C. Prior calibration of the layer (Stage 0)

Before training any manager, the substrate was calibrated with forced arms and paired seeds. (i) *Zero interface tax*: the layer reproduces the MASS script bit for bit and nails the scripted bait's reference on the same 58 two-front pairs (identical severity 58/58): executing the tactic through the interface costs nothing. (ii) *The bait's margin as a decision*: $\Delta(\text{BAIT} - \text{MASS})$ in the grouped stratum grew with each world correction ($+0.43 \to +0.85 \to +1.16$ from v3.4 to v3.6), and the dispersed stratum flipped sign under the static patrol ($\Delta 90$: $-0.44 \to +0.32$), up to the clean v3.7 cells of Sec. V. (iii) *Latencies*: the 75th percentile of the time until the barrier commits (2.3–4.3k ticks) exceeded the pre-registered 2,000-tick threshold, triggering—twice, in two worlds—the rule that chose event-based termination over a fixed clock. (iv) *Quorum frontier*: with $n \leq 2$ wolves, hunting against the real defense is residual (0.15–0.18 kills/episode), so training samples $n \in \{3, 4, 5\}$ while evaluation keeps the full natural distribution.

D. Experimental protocol

All experiments use **paired seeds** (same episode, different policy) with bootstrap intervals (10^4 resamples) over per-seed differences, typically with 100–400 pairs per cell. Before each training we freeze a **pre-registration** with re-measured baseline thresholds and a composite success criterion: *structure* (the policy discriminates the state: bait where it pays, split when a second front appears) plus *non-inferiority* against the distilled oracle ($\delta = 0.15$). Every milestone passes three sign-offs in order: automated assertions, **human viewing** of criterion-selected rendered episodes, and numerical analysis. We additionally log a *censoring* metric—whether the play was actually played—to distinguish a zero from not playing from a zero from playing and failing; this metric uncovered two deadlocks in the hierarchical layer itself (Sec. V-D).

As the attacking oracle we use the rule distilled from prior calibration (BAIT if $n \geq 3$, with *keep* under grouped spawns and $\Delta 90$ under dispersed ones; MASS otherwise); as the defensive benchmark, the proportional allocator. All artifacts (time-stamped pre-registrations, evaluations, event timelines, rendered episodes and checkpoints) are archived with a manifest and hashes, and every world is tagged (v3.4–v3.7): every figure in this article is reproducible from its seed; public repository: https://github.com/berordas/drone-herd-protection.

V. Results

A. Flat RL does not cross the valley

On the attacking side, none of the eight flat campaigns made the bait emerge (Table I). From scratch there is no signal (one or two spontaneous kills in $\approx 2{,}300$ episodes, 2.48M steps); with shaping and warm starts, the best result reaches 0.57–0.72 kills/episode against a scripted floor of 2.77 under the same harness (v2.4.1); and in world v3.4, the residual variants—*initialized* at the programmed bait—never beat the floor (2.17–2.71 vs. 2.68) and reproducibly degrade it: the signature of the bait's rear channel (fraction of never-confirmed killers, KNC) drops from 26.6% to 13% over training. The local gradient at the optimum points away from the tactic. The flat campaigns were not re-run in the final v3.7 world (budget), but the bias direction works against flat learning: the v3.5–v3.7 corrections *widened* the valley (the bait's margin-as-a-decision grew from $+0.43$ to $+1.16$, Sec. IV), so the tactic flat RL failed to find in the older worlds now requires crossing an even longer negative stretch. On the defending side, residual MAPPO improves piloting (Sec. V-C) but front splitting does not emerge in 20M steps (5.7% $\to$ 5.3% of episodes with a split).

Fig. 3 materializes the cause: forcing the tactic through the options layer, the bait's cumulative kills run *below* the mass attack's until tick $\approx 2{,}100$ (v3.5; $\approx 4{,}700$ in v3.4) and only cross in the episode's tail. That negative stretch of thousands of steps is the valley that gradient descent does not traverse and that residual erosion confirms from the other edge.

B. Attacking manager: invariant and non-inferiority

Table II summarizes the comparison ladder in the final world (v3.7), with thresholds frozen by pre-registration before

TABLE I

THE EIGHT FLAT ATTACKING-SIDE CAMPAIGNS.

Runs	Approach	Outcome
01	PPO, sparse reward	no signal (1–2 kills in ≈ 2,300 eps)
02	+ potential shaping	0.57
03	+ behavior cloning (warm start)	0.65–0.72
04	residual on the script	≈ floor; no new tactic
05–06	+ separation curriculum	bait erosion
07–08	+ two-front diet	2.17–2.71 ≤ floor 2.68; KNC 26.6%→13%

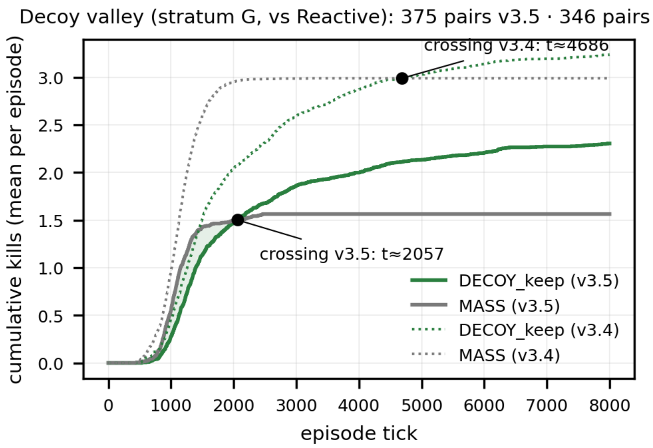

Fig. 3. The measured reward valley: mean cumulative kills of the bait versus the mass attack (grouped stratum, paired seeds). The tactic underperforms for thousands of ticks before overtaking; crossing at ≈ 2,100 in v3.5 (solid) and ≈ 4,700 in v3.4 (dotted).

training. The manager beats every fixed rule and is non-inferior to the distilled oracle, with a point advantage of $+0.10$ $[+0.03, +0.19]$ in the main run; the replication with a different seed reproduces the core (1.69 $[1.38, 2.01]$; $+0.03$ $[-0.12, +0.18]$ against the oracle, within δ).

The central result is not the margin but the **convergence invariant**: in the three independent trainings (main run, fixed-termination ablation and replication), the policy converges to the same choice in the dispersed-spawn stratum, $P(\Delta 90 \,|\, S) = 1.000$ (Fig. 4), matching the best arm of the reward landscape measured in forced cells (MASS 0.30 / Δ90 1.71 $[1.29, 2.14]$ / Δ180 1.42). The manager thus rediscovers, from pure reward and with no rule dictating it, the strategy that prior experimental calibration had distilled by hand. In the grouped stratum, the choice varies across seeds (*keep* in the main run, Δ180 in the ablation and the replication) with no appreciable performance difference between runs, pointing to an approximately flat landscape between the two alternatives: the policy converges exactly where the landscape discriminates, and fluctuates where there is nothing to learn. The same holds in the $n \leq 2$ cells, where the policy extends the bait against the oracle's dictate: hunting there is residual under every arm (0.15–0.18, Sec. IV) and, moreover, the layer's quorum

TABLE II

ATTACKING-SIDE LADDER (LIVESTOCK KILLED/EPISODE; HIGHER IS BETTER FOR THE ATTACKER; 100 PAIRED SEEDS, DEFENSE: CLASSICAL BARRIER V3.7).

Policy	Sev.	Δ vs. oracle [95% CI]
MASS always	0.56	—
Bait iff grouped spawn	0.95	—
Distilled oracle	1.66	—
Manager (main run)	**1.76**	$+0.10$ $[+0.03, +0.19]$
Manager (replication)	1.69	$+0.03$ $[-0.12, +0.18]$

TABLE III

DEFENDING SIDE (LIVESTOCK LOST/EPISODE; LOWER IS BETTER; 100 PAIRED SEEDS PER CELL AND RUN; JOINT ΔS OVER 200 PAIRS).

Defense	Natural mix	Scripted bait	Attacking manager
Classical barrier 4-0	0.74	2.61	1.76
Proportional scripted	0.25	0.85	0.61
Manager (main run)	**0.21**	1.03	**0.67**
Manager (replication)	0.21	1.06	0.76
Joint Δ vs. 4-0	-0.53	-1.56	-1.05
Joint Δ vs. prop.	-0.04 $[-0.10, 0.01]$	$+0.20$ $[0.07, 0.33]$	$+0.10$ $[0.02, 0.19]$

safeguard degrades BAIT to MASS since no assault can form—the choice is literally inconsequential there. The learned policy transfers without degradation to the project's two residual-MAPPO defenses (run02 and run09, trained in worlds v3.4 and v3.5 and frozen), never seen in training (1.75/1.76), and its decision discipline is as designed: 3.5 decisions/episode, the play fully executed in 100% of episodes, residual aborts at 0.28/episode and deliberation cost paid ≈ 0.

The fixed-termination ablation ($K = 1{,}000$ ticks) learned the same structure but with a degraded margin and $\approx \times 3.25$ the compute: event-based termination turned out to be an *ingredient of the margin, not of the mechanism*—the strong pre-registered prediction (collapse) failed, and we report it as such; what crosses the valley is the structural temporal abstraction: that the option exists and executes the entire play. The replication also reproduced the discipline (aborts 0.46/episode, play fully executed in 95.5%), and the ablation kept $P(\Delta 90 \,|\, S) = 1.000$ despite the fixed clock: three independent optimizations, three arrivals at the same point of the discriminating stratum.

C. Defending manager: splitting emerges and reproduces

The defending manager learns conditional splitting and reproduces it in replication: $P(\text{guards} \,|\, 2^{\text{nd}} \text{ cluster}) = 1.000$ and 0.943 in the two runs, with 4-0 dominant under a single front and split latencies of tens of ticks (two defender trainings versus the attacker's three: the termination ablation ran only on the wolf side, for compute-budget reasons). Against the classical barrier it cuts losses in all three conditions with intervals excluding zero (Table III), the bait's rear channel is closed (KNC $= 0$ in the natural and bait cells)[2], and the

[2]KNC is not computed for the manager attacker in Table III: that cell's harness does not log the killer's confirmation status. The wolf manager's KNC measured in its own evaluation (v3.7, vs. the classical barrier) is 0.363 (oracle: 0.404).

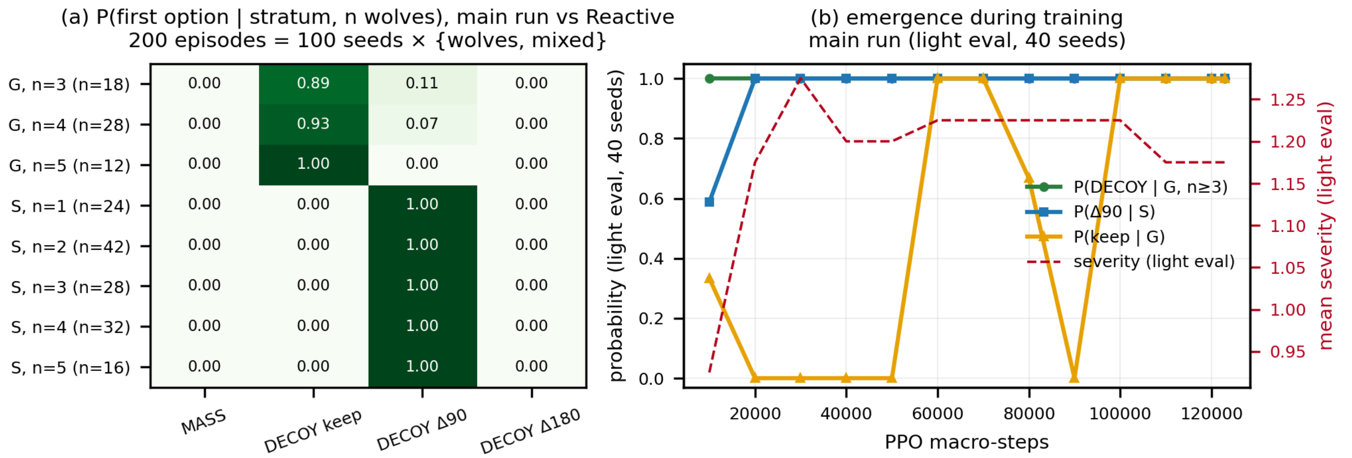

Fig. 4. (a) Final policy of the attacking manager: probability of the first option by spawn stratum and wolf count (200 episodes = 100 paired seeds × two episode types; *keep* under grouped spawns 0.89–1.00; Δ90 under dispersed ones 1.00 for all n). (b) Emergence curve during training (light evaluation, 40 seeds).

guards win the redeployment race in 78–89% of episodes. Against the proportional allocator it is non-inferior on the natural mix (-0.04 $[-0.10, +0.01]$) and loses reproducibly in the other two cells: clearly against the scripted bait ($+0.20$ $[+0.07, +0.33]$ joint) and by a small but statistically significant margin against the adaptive attacker ($+0.10$ $[+0.02, +0.19]$). The reading is consistent: against a rigid attacker that always executes the same pattern, a fixed rule tuned to that pattern remains superior; against the adaptive one—the relevant one—the learned manager stays within a tenth of the best rule, contributing in exchange less flicker and a deployment preference no rule contained. Two unanticipated findings: the manager *almost never* uses the 3-1 deployment foreseen in the design (0.000 and 0.019 across the two runs) and always splits to 2-2, a preference absent from all of our rules; and splitting has a measured price: the two-drone line suffers more penetrations (Fig. 5), a direct consequence of the acoustic geometry (20 m per drone). The manager's discipline matches: 1.2 reassignments/episode versus the proportional rule's 22 (no deployment flicker), split latencies of 10–104 ticks depending on run and seed, a deliberation cost paid of 0.014–0.019, and a pattern of *precautionary guards* in quiet episodes (147–189 of 400 episodes with guards detached without a confirmed second front): an insurance learned against the training distribution—natural mix plus the attacking manager—which in the natural cell costs almost nothing (it is the best figure in that column) and is reported unoptimized.

Comparing the two levels of defensive learning: learned fine piloting (residual MAPPO) improves on the barrier by ~ 0.2 (0.77 vs. 0.93), whereas learned allocation improves on it by 0.5–1.6 depending on the attacker: the margin resided in high-level coordination, not in low-level control.

D. Specification gaming as a process result

The protocol caught three episodes in which the manager optimized interface loopholes instead of tactics, all invisible

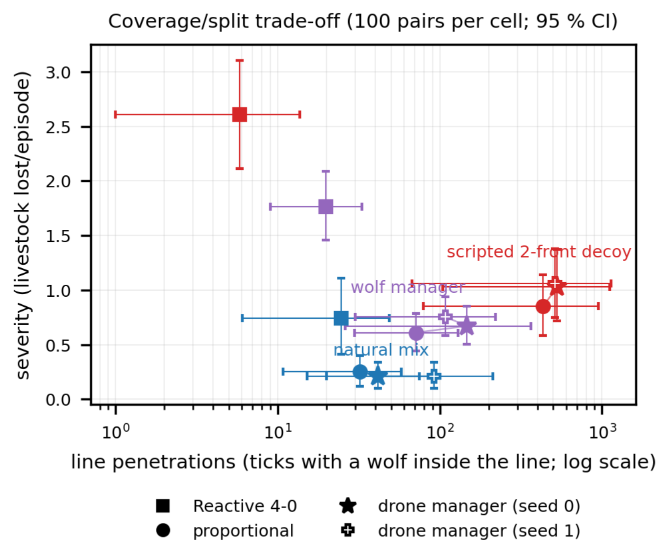

Fig. 5. Coverage/splitting trade-off: line penetrations versus severity, by defense and attacker type. Splitting buys the second front at the price of thinning the first.

to automated tests and discovered by the combination of preregistered gates and human viewing: (i) *prey re-selection*: restarting options re-selected the assault's prey, turning decision churn into a re-targeting wheel the barrier could not follow (an inflated advantage of $+0.42$); it was resolved by legalizing the capability—switching prey is realistic hunting—but regulating it with a hunting condition and granting it to every comparison arm; (ii) *heading windmill*: a mis-specified abort event kept firing after the trigger, allowing the assault to rotate faster than the barrier could re-anchor (45 interruptions/episode); the deliberation cost plus the event fix reduced it to 0.28, and forensic analysis revealed that part of that margin consisted of *unblocking* two deadlocks of the layer itself (a $\pm 25°$ heading requirement and an approach plateau) that left the oracle unable to play in 9 out of 20 dispersed-stratum episodes—in the worst case an entire episode (23,570 ticks) waiting for a geometrically impossible signal—both fixed to best-effort semantics; (iii) the witness episode (seed 21) summarizes the trajectory: 7 kills with the first exploit, 6 with the windmill, 0 under clean play. The lesson generalizes: with a competent optimizer, every interface condition is a rule of the game, and pre-registration with behavioral auditing is what separates "it learned the tactic" from "it found the hole." Three practices proved decisive: turning every visual finding into a regression assertion (the bait's ordering, the closed corridor, no-stacking); leaving passive *tripwires* on accepted defects (counters that flag the episode and send it to human viewing if the optimizer relapses); and keeping a public errata log in the artifact master table (denominators, double roundings, per-seed attributions), so that no number is ever corrected silently.

VI. Discussion and Limitations

Why the hierarchy works. The valley is not crossed by granting more freedom at the movement level—eight campaigns and 20M steps attest to it—but by lifting the coordinated decision to a level where trying it costs one choice: "start baiting" goes from an improbable choreography to one of four discrete actions, and the full option's return makes the delayed payoff visible. Event-based termination further cheapens credit assignment (the payoff falls inside the option that caused it); the ablation shows that without it the structure still emerges, but with a worse margin and triple the compute.

Rigid versus adaptive. The proportional rule clearly beats the manager only against the fixed-pattern attacker (the scripted bait, $+0.20$); against the adaptive attacker the gap shrinks to a tenth, and on the natural mix it vanishes. The reading is consistent: the rule buys specialization on the pattern it knows; learning buys adaptation and contributes preferences contained in no rule, such as the 2-2 split.

Toward the field. The simulator takes an optimistic stance on two points that real-world transfer must revisit: always-effective acoustic deterrence (real predators habituate to static deterrents, suggesting stimulus alternation or reserving sound for close contact) and two-radius perception as a geometric oracle, whose natural substitute is an onboard visual detector; commercial virtual-fencing collars would additionally make active herd escorting plausible. The double benefit that motivates the system—zero losses and, thereby, zero retaliation—depends on these simplifications surviving the field.

Limitations and future work. (1) A single environment with frozen low-level options: joint fine-tuning (e.g., residual policies under the manager) remains future work, with the observed erosion as a documented risk. (2) Managers were trained against frozen opponents; co-evolution of both sides (alternating training with per-phase frozen opponents) is the natural extension, with the known recipe of semi-static phases: each side trains against the other frozen and is evaluated against a fixed suite to avoid cycling. (3) The attacking manager is omniscient by adversary model. (4) The 2-2 penetration trade-off and precautionary guards in quiet episodes are measured but not optimized; the "proportional-2-2" rule distilled from the manager itself is a candidate new benchmark. (5) Strict patrol surveillance coverage is unattainable with 4 drones except at the minimum of the spread range ($D = 2R\sin(\pi/M)$): more units, or anchoring to the most exposed animal, are natural improvements. (6) Numbers come from a single custom simulator; the published code, seeds and artifacts allow exact reproduction (bitwise determinism verified). (7) We did not compare empirically against the directed-exploration or optimism families (Sec. II); the comparison is conceptual and its implementation remains future work.

VII. Conclusion

In a drone-based predator–defense game, the two-front tactic is separated from greedy behavior by a reward valley that flat RL does not cross on either side. A minimal hierarchical manager—PPO over frozen scripted options, an event-terminated semi-MDP with a deliberation cost—crosses it on both: the attacker rediscovers the expert strategy with a convergence invariant in 3/3 trainings and replicated non-inferiority; the defender learns conditional splitting and cuts the classical barrier's losses by a factor of 2.5–3.5, advancing the practical goal that motivates the system: herds without losses and, thereby, wolves without retaliation. The same protocol that produced these numbers caught three episodes of specification gaming and five environment defects before they could contaminate the conclusions: in the presence of an optimizer, methodology is not an accompaniment to the result; it is part of the result.

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